

Chapter One

General Pharmacology

Learning Objectives

- Up on completion of this course, students will be able to
 - Understand the general principles of pharmacology.
 - Explain the pharmacokinetics of drugs affecting the nervous system, respiratory system, gastrointestinal system and autacoids.
 - Explain the pharmacological actions, mechanism of actions, and therapeutic uses and pharmacokinetics of drugs affecting the nervous system, respiratory system, gastrointestinal system and autacoids.
 - Explain side effects of drugs acting on autonomic nervous system, central nervous system, respiratory, GI, and drugs acting on inflammation.
 - Apply concept and principles of pharmacology to ensure and proper use of drugs.
 - To work in Pharmacology Laboratory and will be able to practice selected basic experimental demonstration

Introduction

- ❖ Pharmacology is the science of how **drugs** act on biological systems and how the body responds to the drug.
- ❖ the study of substances that interact with living systems through chemical processes
- ❖ These substances may be
 - ✓ chemicals administered to achieve a beneficial therapeutic effect on some process within the patient or
 - ✓ their toxic effects on regulatory processes in parasites infecting the patient

Basic areas of pharmacology

- **pharmacokinetics:** deals with absorption, distribution, metabolism and excretion of drugs
- **pharmacodynamics:** the study of biochemical and physiological effects of drugs and their MOA
- **Pharmacotherapy:** the science of substances used to prevent, diagnose, and treat disease
- **Toxicology** is the branch of pharmacology that deals with the undesirable effects of chemicals on living systems.
- **Pharmacogenomics:** relationship between individual genetic makeup and his/her response to specific drugs

- ❖ The study of pharmacology encompasses
 - the sources,
 - chemical properties,
 - biological effects and therapeutic uses of drugs.

Historical perspectives

❖ In the last 1500 years

- ✓ Attempt was done to explain all of **biology and disease without the need for experimentation and observation**

Assumption = disease was caused by **excesses of bile or blood in the body**

Around the end of the 17th century

- ✓ **observation and experimentation** began to replace theorizing in **physiology and clinical medicine**
- ✓ **materia medica** was developed

Historical perspectives

- ❖ **In the late 18th and early 19th centuries**
- ✓ **François Magendie** and his student **Claude Bernard** began to develop the **methods of experimental physiology and pharmacology**
- ✓ Advances in chemistry and the further development of physiology in the 18th, 19th, and early 20th centuries laid the foundation needed for understanding **how drugs work at the organ and tissue levels.**
- ❖ **1940s and 1950s**
- ✓ Information accumulated about drug action and the **biologic substrate of that action, the drug receptor**

Drug, definitions and source

Drug

- ❖ Is any substance or product used or intended to be used to **modify** or explore the physiological state or **pathological condition** for the well being of the recipient (WHO)
- ❖ Any chemical agent that affects living protoplasm
- ❖ any substance (other than food that provides nutritional support) that, when
 - ✓ inhaled,
 - ✓ injected,
 - ✓ smoked, consumed
 - ✓ absorbed via a patch on the skin, or dissolved under the tongue causes a temporary **physiological** change in the body.

Sources

❖ 1. natural drugs

❖ **Plants;** Roots, bark, sap, leaves, flowers, seeds were sources for drugs e.g.

- Reserpine from Rauwolfia Vomitoria,

- Digitalis from foxglove,

- opium from the poppy plant.

❖ **Animal sources;** gave us hormones for replacement in times of deficiencies e.g. Insulin from the **pancreases of pigs** and cattle, Liver extracts for anemia etc

❖ **Minerals:** Iodine, iron, potassium salts.

❖ **Micro-organisms:** penicillin,

2. Synthetic drugs; from natural products in the laboratory.

3. Semisynthetic drugs: by chemical modifications of natural products

4. Biosynthetic drugs: by rDNA technology.

Drug naming system

Proprietary medicines or brand medicines

- Protected by a patent,
the patent usually covers **the formula** and manufacturing process and is usually in force for several years.
- During this time the medicines can not copied legally
- Proprietary medicines are also sold as **brand medicines**, original medicines, innovator products
- **Once the patent expires** the protected information will be part of the public domain i.e the **information** can be used by everybody.
- At this point the medicines may be used as **models** for manufacturing of generic products

International non proprietary name

- Often simply called generics. These are medicines that are copies of proprietary medicines, they contain the **same active ingredient**, at the **same dosage**, and have the **same pharmaceuticals form**.
- They can be manufactured and sold only once the **patents protecting** the proprietary medicines expired.
- Generic medicines are **cheaper** than proprietary medicines.

Chemical Name/Scientific name

-long name, refers to the chemical structure of the drug

N-acetyl-p-aminophenol For PCM

Drug categories...

a) Prescription Drugs

–Prescription drugs are drugs that the federal government has designated to be **potentially harmful** unless their use is supervised by a licensed health care provider, such as a nurse practitioner, physician, or dentist

Drug categories...

b) Nonprescription Drugs

- Drugs that are designated by the federal government to be safe (**if taken as directed**) & obtained without a prescription
- They are also referred to as **over-the-counter (OTC) drugs** & may be purchased in a variety of settings, such as a pharmacy, drugstore, etc
- They include those given for symptoms of the
 - common cold,
 - headaches,
 - constipation,
 - diarrhea, &
 - stomach upset

Drug categories...

c) Controlled Substances

- They are the most carefully monitored of all drugs
- These drugs have a **high potential for abuse** & may cause physical or psychological dependence

D. Pregnancy categories

- Medications are tested to determine the medication's effect on a fetus
- Based on these test results, medications are categorized according to their safety for being administered to pts during pregnancy

Pregnancy categories...

- **Category A:**
 - Adequate & well-controlled studies indicate **no risk** to the fetus in the first trimester of pregnancy or later
 - **levothyroxine, folic acid,**
- **Category B:**
 - Animal reproduction studies indicate **no risk to the fetus**; however, there are no well-controlled studies in pregnant women
 - Example: **metformin, hydrochlorothiazide, , amoxicillin**

Pregnancy categories...

- **Category C:**
 - Animal reproduction studies have reported adverse effects on the fetus; however,
 - there are no well-controlled studies in humans, **but potential benefits may indicate use of the drug in pregnant women despite potential risks**
 - **Example :** Fluconazole, Metoprolol, Sertraline

Pregnancy categories...

- **Category D:**
 - Positive human fetal risk has been reported from investigational or marketing experience, or human studies
 - Considering potential benefit versus risk may, in selected cases, warrant the use of these drugs in pregnant women
 - E.g **Phenytoin, Lisinopril**

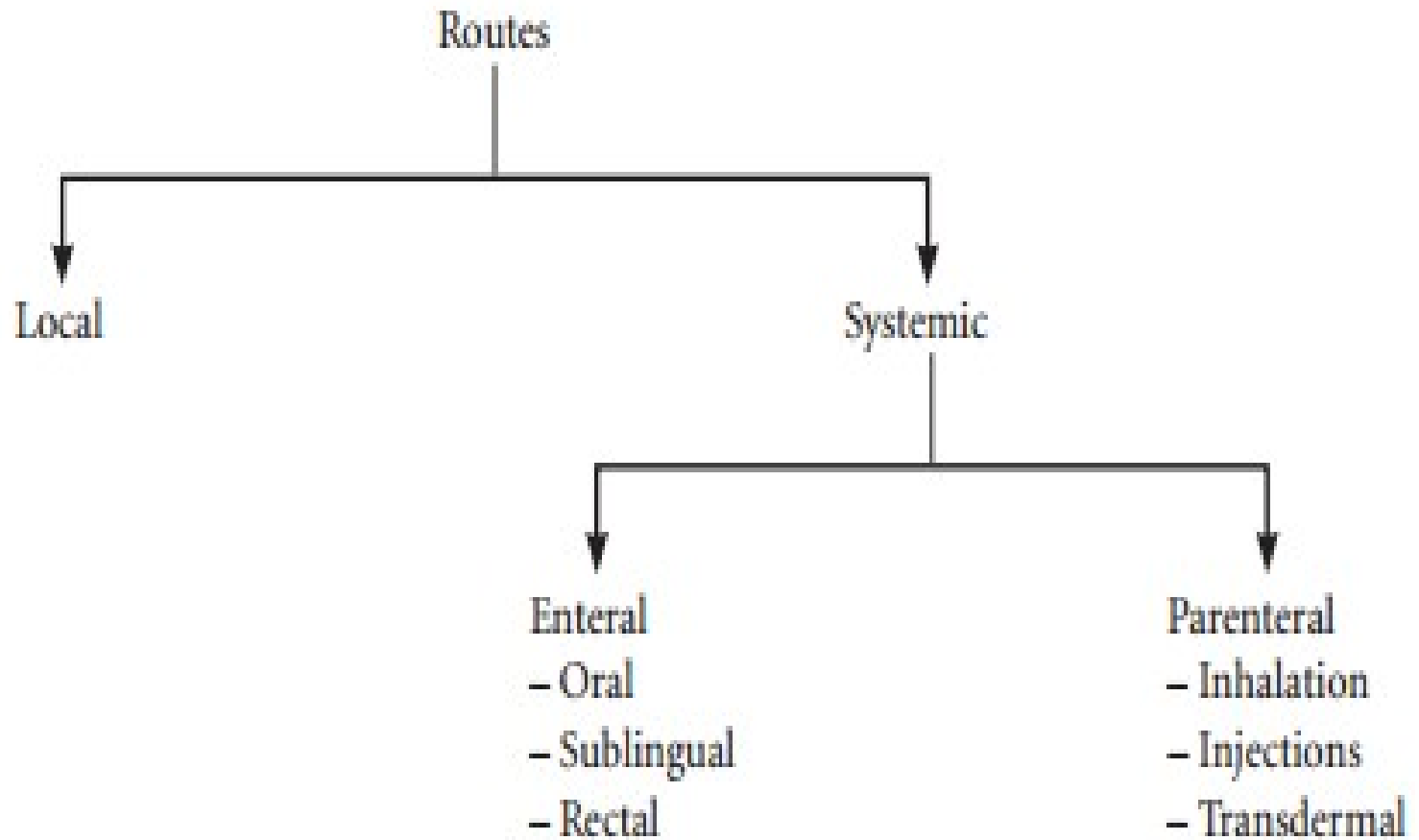
Pregnancy categories...

- **Category X:**

- Fetal abnormalities reported & positive evidence of fetal risk in humans is available from animal &/or human studies.
- The risks involved clearly outweigh the potential benefits
 - These drugs should not be administered to pregnant women

E.g Methotrexate, Warfarin

Routes of drug administration



Local Routes

- ❖ It is the simplest mode of administration of a drug at the site where the desired action is required.
- ❖ Systemic side **effects are minimal**.

Local routes can be delivered by the following

- ❖ **Oral cavity:** As a suspension, e.g. nystatin; as or as oral Jel. clotrimazole (for oral candidiasis);
- ❖ **GI tract:** As tablet that is not absorbed, e.g. neomycin (for sterilization of gut before surgery)

Local Routes

- ❖ **Eye, ear and nose:** As drops, ointments and sprays (for infection, allergic conditions, etc.), e.g. gentamicin eye/ear drops.
- ❖ **Bronchi:** As inhalation, e.g. salbutamol, ipratropium bromide, etc. for bronchial asthma
- ❖ **Skin:** As ointment, cream, lotion or powder, e.g. clotrimazole (antifungal) for cutaneous candidiasis.

Enteral Routes

- ❖ It includes oral, sublingual and rectal routes
- ❖ **Oral Route**
 - ✓ It is the most common and acceptable route for drug administration.
 - ✓ Dosage forms are tablet, capsule, syrup, mixture, etc., e.g., **paracetamol tablet for fever, omeprazole capsule for peptic ulcer** are given orally

Advantages

- Safer. cheaper, painless. convenient for repeated and prolonged use. can be self-administered.

Dis advantage

- Not suitable for **emergency** as onset of action of orally administered drugs is slow.

❖ Sublingual Route

- The preparation is kept under the tongue.
- The drug is absorbed through the buccal mucous membrane and enters the systemic circulation directly, e.g. **nitroglycerin** for **acute anginal** attack and **buprenorphine** for **myocardial infarction**
- **Advantages**
 - Quick onset of action.
 - Action can be terminated by spitting out the tablet.
 - Bypasses first-pass metabolism.
 - Self-administration is possible.

❖ **Sublingual Route...**

➤ **Disadvantage...**It is not suitable for:

- Irritant and lipid-insoluble drugs.
- Drugs with bad smell and taste.

❖ **Rectal Route**

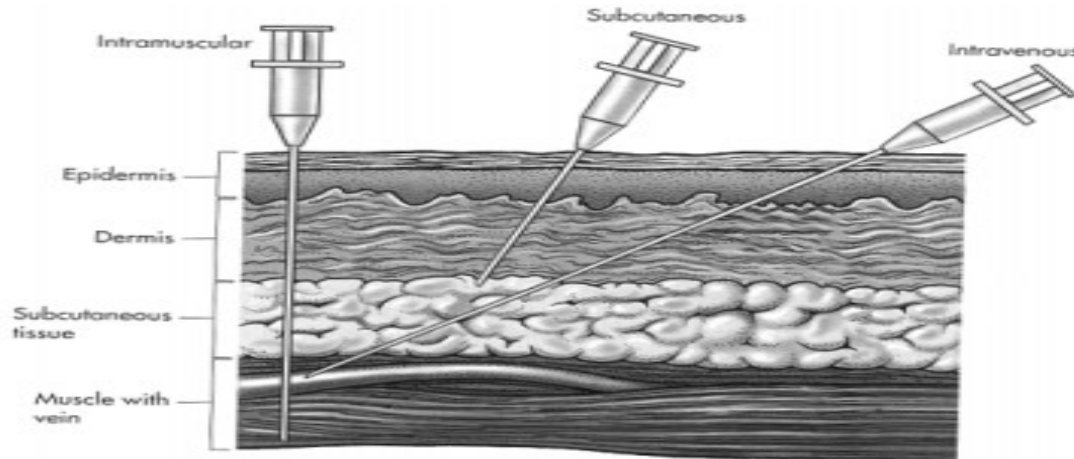
Drugs can be given in the form of solid or liquid.

- **Suppository:** It can be used for local (topical) effect (as well as systemic effect, e.g. **indomethacin** for **rheumatoid arthritis**).
- **Enema:** Retention enema can be used for local effect as well as systemic effect.

Parenteral Routes

- Routes of administration other than enteral route are called **parenteral routes**
- **Advantages**
 - ✓ Onset of action of drugs is faster; hence it is suitable for emergency.
 - ✓ Can be for Unconscious patient.
 - ✓ Uncooperative patients.
 - ✓ Patients with vomiting and diarrhea.
- **Disadvantages**
 - ✓ Require aseptic conditions.
 - ✓ Preparations should be sterile and is expensive.
 - ✓ Requires invasive techniques that are painful.
 - ✓ Cannot be usually self-administered.
 - ✓ Can cause local tissue injury to nerves, vessels, etc.

Injection sites



Subcutaneous (s.c.) route

- The drug is injected into the subcutaneous tissues of the thigh, abdomen and arm, e.g. adrenaline, insulin, etc
- **Advantages**
 - ✓ Self-administration is possible (e.g. insulin).
 - ✓ Depot preparations can be inserted into the subcutaneous tissue, e.g. norplant for contraception.

Disadvantages

- ✓ It is suitable only for nonirritant drugs.
- ✓ Drug absorption is slow; hence it is not suitable for emergency.

Intramuscular (i.m.) route: Drugs are injected into large muscles such as deltoid, gluteus maximus and vastus lateralis,

e.g., diclofenac, etc. A volume of 5–10 mL can be given at a time.

Advantages

- ✓ Absorption is more rapid as compared to oral route.
- ✓ Mild irritants, depot injections, soluble substances and suspensions can be given by this route.

Disadvantages

- ✓ Aseptic conditions are needed.
- ✓ Intramuscular injections are painful and may cause abscess.
- ✓ Self-administration is not possible.
- ✓ There may be injury to the nerves.

Intravenous (i.v.) route: Drugs are injected directly into the blood stream through a vein.

Drugs can be given as

Bolus: Single, relatively large dose of a drug injected rapidly or slowly as a single unit into a vein. For example, i.v. ranitidine in bleeding peptic ulcer

Slow Intravenous infusion: For example, dopamine infusion in cardiogenic shock

Advantage

- ✓ Bioavailability is 100%
- ✓ Quick onset of action
- ✓ Large volume of fluid can be administered

✓ **Disadvantage**

- ✓ Once the drug is injected, its action cannot be halted.
- ✓ Self-medication is not possible.
- ✓ Strict aseptic conditions are needed.
- ✓ Extravasation of some drugs can cause injury, necrosis and sloughing of tissues.

Mechanisms of drug permeation

- ❖ Drug permeation proceeds by several mechanisms.
- ✓ **Passive diffusion** in an aqueous or lipid medium is common,
- ✓ **active processes** play a role in the movement of many drugs, especially those whose molecules are too large to diffuse readily

Aqueous diffusion: Aqueous diffusion of drug molecules is usually driven by the concentration gradient of the permeating drug

- permit the passage of molecules as large as MW 20,000–30,000 Da

Mechanisms of drug permeation...

- ❖ Factors affecting diffusion
 - ❖ bound to large plasma protein
 - ❖ Charging of the drug
- ❖ The passive flux of molecules down a concentration gradient is given by Fick's law

Flux (molecules per unit time) =

$$(C_1 - C_2) \times \frac{\text{Area} \times \text{Permeability coefficient}}{\text{Thickness}}$$

Active transport

Active transport

- ✓ is the movement of molecules across a membrane from a region of lower concentration to a region of higher concentration—against the concentration gradient.
- ✓ Active transport requires cellular energy to achieve this movement

❖ *Special carriers* _ Special carrier molecules exist for many substances that are important for cell function and too large or *too insoluble* in lipid to diffuse passively through membranes, eg, **peptides, amino acids, and glucose**

Special carriers...

- ❖ movement is through by *active transport* or **facilitated diffusion**
- ❖ They are selective and **saturable**

Endocytosis and exocytosis _ A few substances are so **large** or **impermeant** that they can enter cells only by endocytosis,

- ✓ In endocytosis the substance is bound at a **cell surface** receptor, **engulfed by the cell membrane**, and carried into the cell by pinching off of the newly formed vesicle inside the membrane.
- ✓ The substance can then be released inside the cytosol by breakdown of the vesicle membrane

E.g Vit B complex and iron

- ✓ Exocytosis is the reverse process
- ✓ for the secretion of many substances from cells

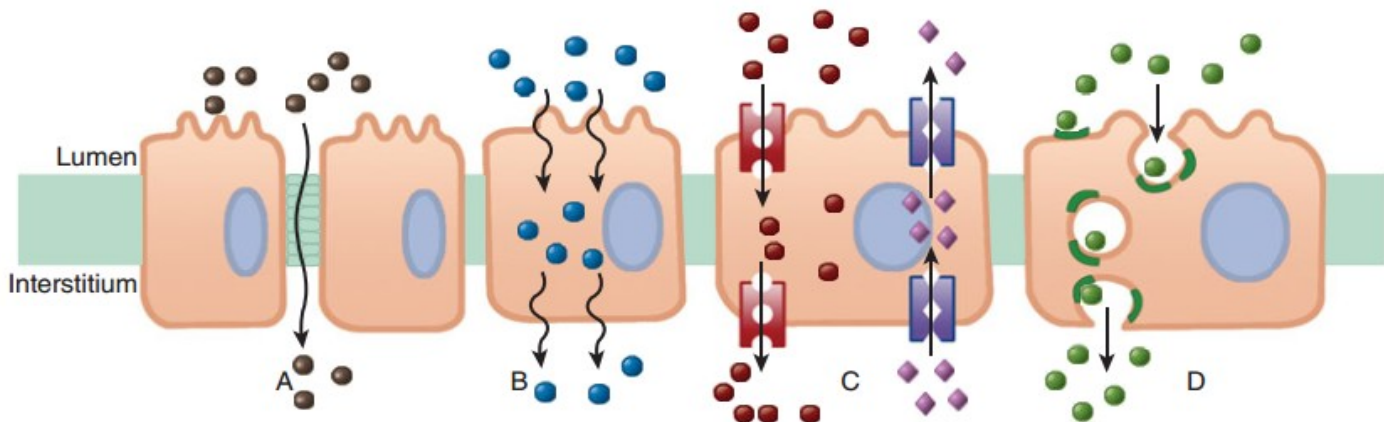


fig : mechanisms of drug permeation

Keys: a: passive diffusion

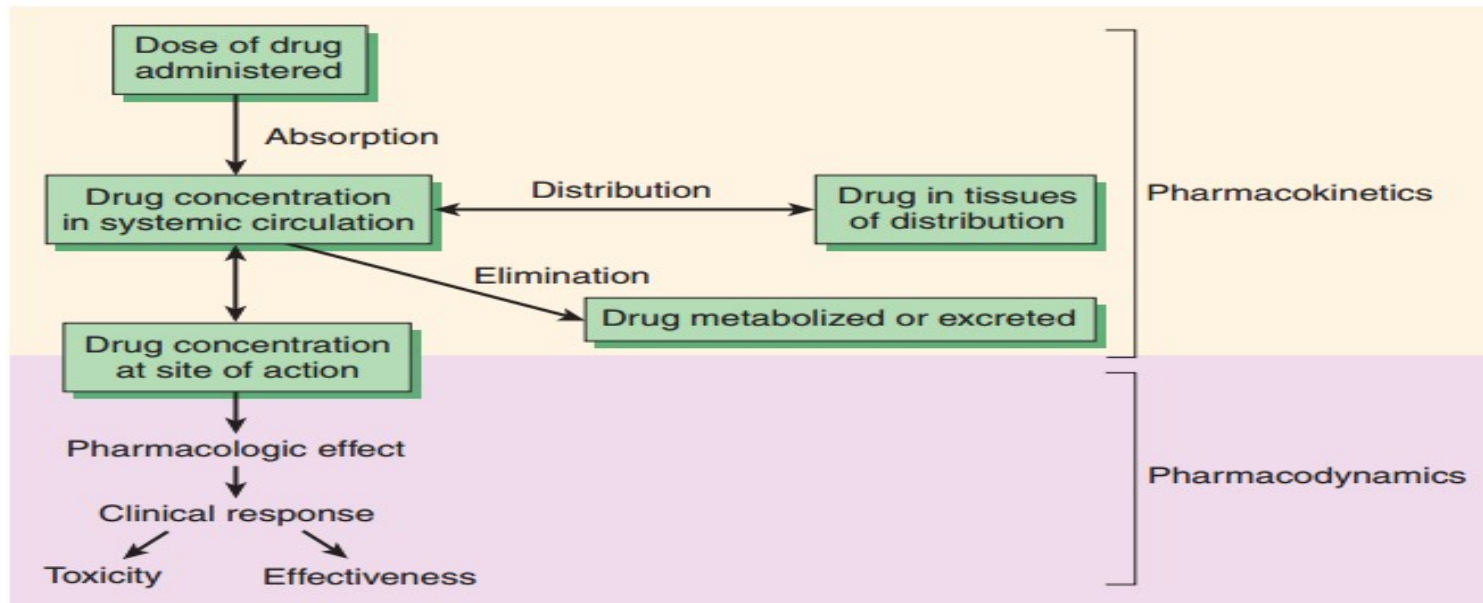
c: endocytosis

b: Facilitated diffusion

d: exocytosis

Pharmacokinetics

- Pharmacokinetics (PK) deals with
- drug absorption, distribution, metabolism and excretion (ADME)



Absorption

- ❖ **Absorption** is the movement of a drug from its site of administration into the **central compartment**.
- ❖ For **solid dosage** forms, *absorption* first requires dissolution of the tablet or capsule, thus liberating the drug
- ❖ **Bioavailability** describes the fractional extent to which a dose of drug reaches its site of action or a biological fluid.
- ❖ **Factors affecting drug absorption**
 - Patient related factors
 - Pharmaceutical factors

Patient related factors

❖ Patient factors is related to the following

➤ Gastric emptying rate

Drug swallowed-----reaches to stomach.....small intestine (duodenum), if a delay in release of gastric content to small intestine.....***absorption delays....irritation/ instability will happen***

✓ Gastric emptying rate is faster in case of **solution & suspensions** than solid & non disintegrating dosage forms

➤ Intestinal motility

➤ Normal peristaltic movements..... drug will have a sufficient time (*residence time*) at the absorption site for optimum absorption.

➤ **Blood flow to GIT...greater blood flow greater absorption**

Patient related factors...

- ❖ Effect of Food... it may increase or decrease
- ❖ penicillin and tetracycline..... ↓ griseofulvin↑

Pharmaceutical factors

Pharmaceutical factors is related to

- drug solubility & dissolution rate
- Particles size & effective surface area
- Solvates & hydrates
- salt form of drug
- Ionization state

Distribution of Drug

- ❖ ***Distribution*** is a movement of drug to site of ***action***
 - a drug distributes into **interstitial** and **intracellular fluids** as functions of the physicochemical properties of the drug,
 - the rate of drug delivery is at different capacities at different regions to interact with the drug.
 - Cardiac output,
 - regional blood flow,

- ❖ well-perfused organs
 - liver,
 - kidney,
 - Brain.
- ❖ Less perfused organs
 - muscle,
 - most viscera, skin, and
 - fat

Distribution of Drug...

❖ well-perfused organs

- liver,
- kidney,
- Brain.

❖ Less perfused organs

- muscle,
- most viscera, skin, and
- fat

Distribution of Drug...

Binding to Plasma Proteins.... Many drugs circulate in the bloodstream bound to plasma proteins.

- Albumin is a major carrier for acidic drugs;
- α 1-acid glycoprotein binds basic drugs.
- Nonspecific binding to other plasma proteins generally occurs to a much smaller extent.
- The binding is usually **reversible**.

Distribution of Drug...

Tissue Binding

- ❖ Many drugs accumulate **in tissues** at higher concentrations than those in the extracellular fluids and blood.
- ❖ Will serve as a **reservoir** that prolongs drug action in
- ❖ accumulation also can produce **local toxicity** (e.g., **renal** and **ototoxicity** associated with aminoglycoside antibiotics).

Distribution of Drug...

CNS, the BBB, and CSF

- The brain capillary endothelial cells have continuous **tight junctions**; therefore, drug penetration into the brain depends on **transcellular** rather than **paracellular** transport.
- The **lipid solubility** of the nonionized and unbound species of a drug is an important determinant of its uptake by the brain;
- The more lipophilic a drug, the more likely it is to cross the BBB.

Distribution of Drug...

Bone

- **Bone** is a reservoir for the slow release of toxic agents such as **lead** or **radium**;
- their effects thus can persist long after exposure.
- **Local destruction** of the bone medulla also may result in reduced blood flow and prolongation of the reservoir effect as the toxic agent becomes **sealed off** from the **circulation**; this may further enhance the direct local damage to the bone.

Distribution of Drug...

Fat

- Fat will serve as a **reservoir**
- **Many lipid-soluble** drugs are stored by physical solution in the neutral fat.
- In obese persons, the fat content of the body may be as high as **50%**, and even in lean individuals, fat constitutes **10%** of body weight; hence, fat may serve as a reservoir for lipid-soluble drugs.
- Fat is a rather **stable reservoir** because it has a relatively low blood flow.

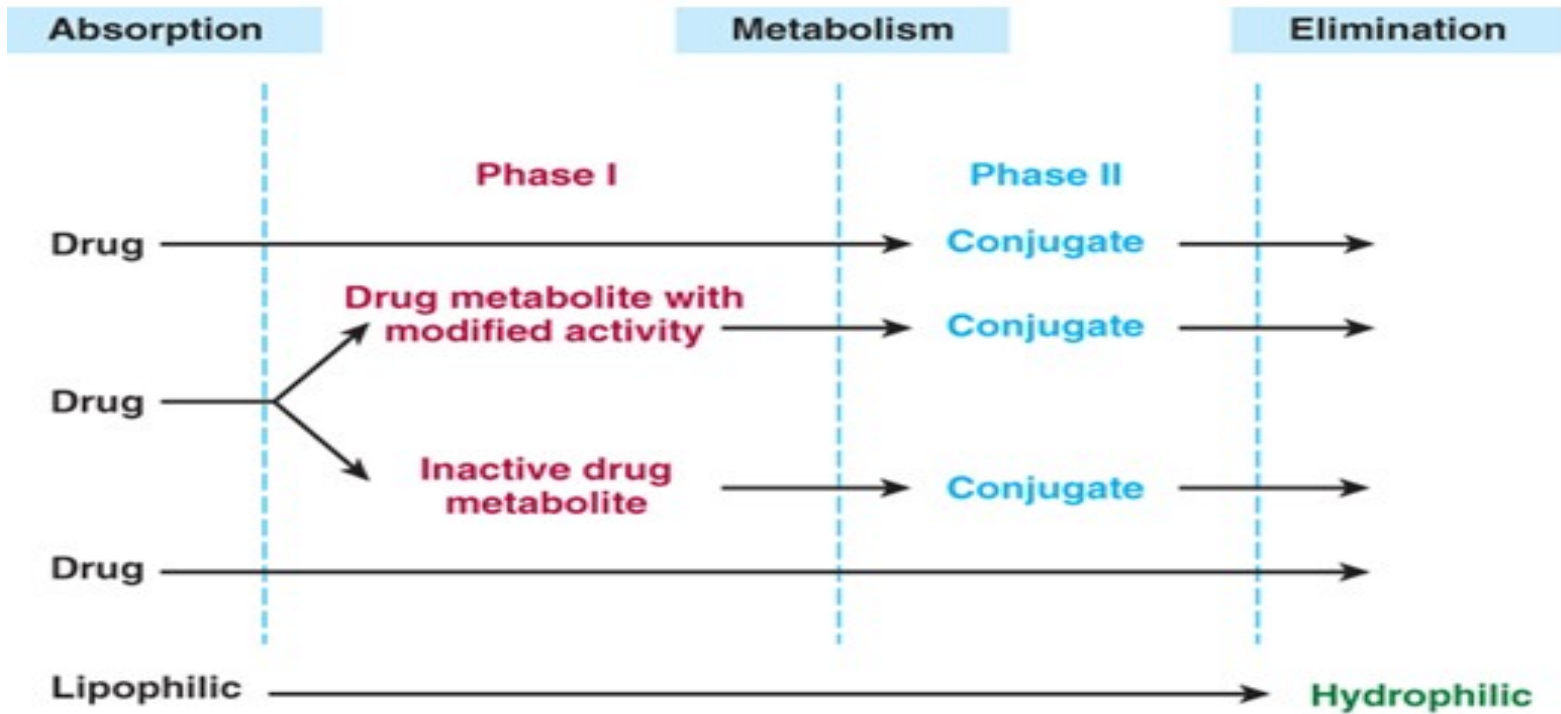
Metabolism

- ❖ The metabolism of drugs and other xenobiotics into more hydrophilic metabolites is
 - essential for the elimination of these compounds from the body and termination of their biological activity.

First-Pass Metabolism

- Following **non parenteral** administration of a drug,
- a significant portion of the dose may be metabolically inactivated in either the intestinal **endothelium** or the **liver** before it reaches the **systemic circulation**
- Limits oral availability of highly **metabolized drugs**

Metabolism



Biotransformation

- ✓ Generates more polar (water soluble), inactive metabolites
- ✓ Readily **excreted from body**
- ✓ **Metabolites** may still have potent biological activity (or may have toxic properties)
- ✓ Generally applicable to metabolism of all **xenobiotics** as well as **endogenous compounds**

Phase I and Phase II Metabolism

- Phase I
 - functionalization reactions
- Phase II
 - conjugation reactions

Phase I

- ❖ Converts the parent drug to a more polar metabolite by introducing or unmasking a functional group (-OH, -NH₂, -SH).
- ❖ Usually results in **loss** of pharmacological activity
- ❖ **Sometimes may** be equally or more active than parent

Phase I...

- ✓ Includes **oxidation, reduction, hydrolysis**, and hydration and isomerization
- ✓ Many drugs undergo a **number of these reactions**
- ✓ Main function of Phase I metabolism is to prepare the compound for phase **II metabolism**
- ✓ Mixed function enzyme system found in microsomes of many cells (esp **liver, kidney, lung, intestine**) performs many different functionalization reactions

Phase II Metabolism

- ❖ Phase II is usually the true detoxification of drugs
- ❖ Occurs mostly in cytosol
- ❖ Gives products that are generally water soluble and easily excreted

Includes sugar conjugation, sulfation, methylation, acetylation, amino acid conjugation, glutathione conjugation

Phase II (conjugation reactions)...

- ✓ Subsequent reaction in which a covalent linkage is formed between a functional group on the parent compound or Phase I metabolite and an endogenous substrate such as *glucuronic acid*, **sulfate**, **acetate**, or an **amino acid**
- ✓ **Highly polar** – rapidly excreted in **urine and feces**
- ✓ **Usually inactive** - notable exception is **morphine**

Site of Biotransformation

- ✓ Enzymatic in nature
- ✓ Enzyme systems involved are localized in **liver**
- ✓ Every tissue has some metabolic activity
- ✓ Other organs with significant metabolic capacity are **GI tract, kidneys and lung**

Excretions of drugs

- ❖ Drugs are eliminated from the body either **unchanged** or as **metabolites**.
- ❖ Excretory organs, eliminate **polar compounds** more efficiently than substances with high lipid solubility.
- ❖ The **kidney** is the most important organ for excreting drugs and their metabolites.
- ❖ Renal excretion of unchanged drug is a major route of elimination for **25-30%** of drugs administered to humans..

Excretions of drugs

- ❖ **breast milk** is important not because of the amounts eliminated, but because the excreted drugs may affect the nursing infant.
- ❖ Excretion from the **lung** is important mainly for the elimination of **anesthetic gases**

Excretions of drugs

Renal excretion

- ❖ Excretion of drugs and metabolites in the urine involves 3 distinct processes: **glomerular filtration**, **active tubular secretion**, and **passive tubular reabsorption**.
- ❖ Changes in overall renal function generally affect all 3 processes to a similar extent.
- ❖ In **neonates**, **renal function** is low compared with body mass but matures rapidly within the first few months after birth.
- ❖ During adulthood, there is a slow decline in renal function, $\sim 1\%$ per year, so that in elderly patients a **substantial degree of functional impairment** may be present.